

DMV302 Data Analysis and Pre-Processing

‘Model Report + Data Analysis’



Component code and name	DMV302 Data Mining and Visualization
Assignment name	Data Analytics—Report Data cleaning
Weighting	25%
Total Marks	100 marks
Submission deadline	Week 7
Week issued	Week 3

Brief

In this assessment, you will undertake two tasks, firstly a case study and write a report advising a client of any potential data mining applications they can employ in their business. For this, write a report that is 1,000 words in length (+/-10%). Secondly, pre-process and explore the dataset. You will need to undertake this assessment individually.

Please refer to the Task Instructions (below) for details on how to complete this task.

Instructions

Task 1:

This task is based on a case study. Read the given case scenario and address all the following tasks in a report.

- **General Description:** Provide some data mining processes that Woolworths currently employs.
- **Data mining applications:** List at least three uses of data mining that Woolworth or its store owners can employ to help their businesses operate and grow. For each of the applications, you need to identify:
 - The type of data mining algorithm or a category of data mining algorithm (e.g., clustering, association analysis, classification, regression analysis, anomaly detection, or text mining).
 - For each application, briefly state how the business will benefit from the data mining application from either Woolworth's or the store owner's perspective. Your description of the data mining application should be connected to the case study.
- **Computing environment:** Identify key components of the environment, including software, hardware, and analytical tools, for each data mining application that you identified.
- **Data:** Describe in detail the type of data need to be used for each data mining application you identified. Your description should be at a granular level; 'transactional data' is not an acceptable answer, as it fails to explicitly clarify what 'transactional data' encompasses.
- **Challenges:** Discuss the challenges in collecting, preparing, and analyzing the data for the corresponding data mining algorithms. Describe typical data preparation techniques to overcome the challenges in data used for the mining applications that you identified and any other associated data you described. You can assume that Woolworths can collect the data as you described in the normal course of business.

Task 2:

In this task, you are provided with a noisy dataset that consists of attributes of different data types. Your task is to clean the data to make it usable for ML or Mining tasks. You will learn how to read different types of data, handle finding missing values, to apply transformation and encoding methods. Implement the following steps to clean the dataset by applying each operation and saving the transformed attributes separately in data frames.

1. Data Exploration

- a. Data Analysis
- b. Data Visualization and display outliers using suitable graphs.

2. Data Cleaning

- a. Separate the numerical, categorical, text, spatial coordinates, and times-series attributes and save them into different data frames using suitable names.

- b. Detect missing data from each data frame.
- c. Handle missing values for each data frame separately by applying appropriate data handling techniques. Such as dropping attributes with a high ratio of missing values or using filling methods).
- d. Pre-process the text attributes and correct inconsistent data by applying NLP operations.

3. Data Transformation

- a. Transform the time series attributes into different features of time by parsing the dates columns saved above.
- b. Transform the Categorical attributes using different encoding methods
- c. Transform the numerical data using scaling and normalization methods.
- d. Transform text attributes using word vectorization encoding methods.
- e. Transform spatial attributes using appropriate methods.

4. Data Aggregation

- a. Aggregate the different types of processed data frames and display them as the final transformed single dataset.

General assessment instructions

This assessment provides students with an opportunity to demonstrate their knowledge and skills to achieve the required standard. To do this, assessment responses need to be both clear and easy to understand.

Referencing

Students must use the APA style for citing and referencing research appropriately. Please see more information on referencing at <https://mediadesignschool.libguides.com/apa>.

Submission Guidelines

Naming conventions

DMV302LastNameFirstNameBrief1.doc/docx
DMV302LastNameFirstNameBrief1.ipynb

Submission structure

Submit the 1000-word report file via the Assessment 1 link in the main navigation menu in DMV302: Assignment 1, by the end of Week 7.

Assessment Criteria

The brief outline of assessment criteria is as follows. For more details on each section, check the assessment rubrics on MyLearn.

1. Case study analysis (30%)

- Excellent knowledge and analysis of key data mining applications and processes.
- Excellent knowledge and understanding of the types of data used in typical data mining applications and data preparation techniques.
- Analysis and application with the synthesis of new knowledge.

2. Effective communication & References (written) (20%)

- Communicates eloquently. Expresses meaning coherently, concisely, and creatively within the given format.
- Spelling, grammar, and punctuation are free of errors.
- Demonstrates use of high-quality, credible, and relevant resources to support and develop arguments and position statements.
- APA referencing is free of errors

3. Data Cleaning (40%)

- Efficiently fixing or removing incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data within a dataset by applying suitable data cleaning methods.

4. Programming Practices (10%)

- Correct submission of source code (.ipynb) runs with no errors.
- Coding standards fully followed including naming conventions, comments, etc.

Please refer to MyLearn for further details on Rubrics.

Submission Policy

Submit this task via the Assessment 1 link in the main navigation menu in DMV302 Data Mining and Visualisation. The Learning Facilitator will provide feedback via the Grade Centre in the MyLearn portal. You are required to submit ONE Microsoft Word document (.doc or .docx) for task 1 and a Notebook for task 2.

Please ensure that you zip individual files into zip into one folder. Before submitting your assignment, you should check it against the detailed assessment criteria to ensure that you have satisfactorily addressed all the criteria that will be used to mark your assignment.

Please refer to the Submission Policy on MyLearn for further details.

Academic Integrity

I declare that except where I have referenced, the work I am submitting for this assessment task is my work.

I have read and am aware of Media Design School's Academic Integrity Policy and Procedure viewable online at the <https://www.mediadesignschool.com/policies-and-procedures>.

Your work should be a demonstration of your original work and not have any plagiarism issues.